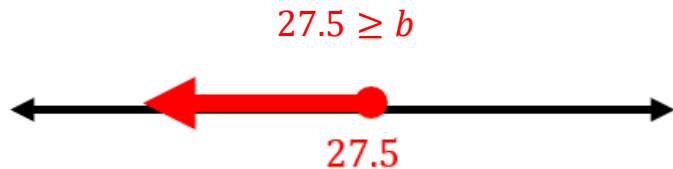
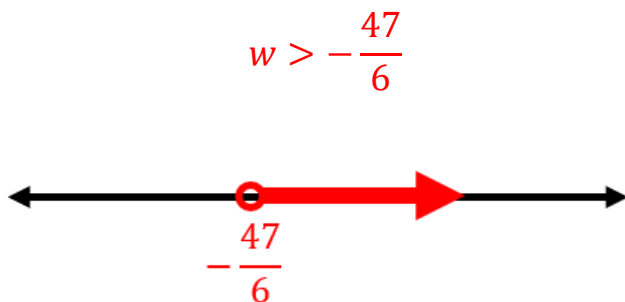


7th Grade Unit 10 Assessment Guide

General Information	Fluency of operations with rational numbers is an important outcome of grade 7. If a calculator is not provided for this assessment, then teachers should consider how to monitor simple calculation errors. If teachers want to consider student fluency of rational numbers, then if a student makes a simple calculation error that is carried through their work, the students receive partial credit. Students may inadvertently write the wrong variable for an answer. This is a common mistake and should receive a minimal point deduction. It is suggested that if a student does it more than twice, 1 point is deducted for each occurrence past two.
Calculator Information	The questions on the unit assessment are calculator neutral, which means that a calculator does not give a student an advantage or disadvantage when showing their knowledge of the intended benchmark.
Total Recommended Points	The total recommended points in this guide is 97 points. Teachers can either determine where best to use the extra 3 points or round the total value up to 100 points.

Question	1	Benchmark(s)	MA.7.AR.2.1
$x > -21.6$ 	Recommended Scoring (8 Total Points)	Suggested points are 4 points for solving the inequality, 2 points for the open circle, and 2 points for the correct ray. Students may write -21.6 using equivalent values.	
	Additional Notes	This is the only question where reversing the inequality sign is assessed. This is because if students fail to do that once it is likely they will repeat the same error. Grade 7 students should be able to distinguish when to reverse the sign. Students who reverse the inequality sign on questions 1 and 4 have a misunderstanding about when to reverse the sign. Students may benefit from reviewing MA.6.AR.2.1 in grade 6 unit 11.	

Question	2	Benchmark(s)	MA.7.AR.2.1
	Recommended Scoring (8 Total Points)	Suggested points are 4 points for solving the inequality, 2 points for the closed circle, and 2 points for the correct ray.	
	Additional Notes	Students may correctly solve the inequality but draw the ray in the incorrect direction. Questions 2 and 4 will indicated to the teacher if the student has this misconception. Students with this misconception may benefit from MA.6.AR.2.1 in grade 6 unit 11.	
Question	3	Benchmark(s)	MA.7.AR.2.1
	Recommended Scoring (8 Total Points)	Suggested points are 4 points for solving the inequality, 2 points for the open circle, and 2 points for the correct ray. Students may write $-\frac{47}{6}$ using equivalent values.	
	Additional Notes	Students may make calculation errors, which may point to a fluency issue with rational numbers. If students express $-\frac{47}{6}$, they should use proper repeating notation to receive full credit.	

Question	4	Benchmark(s)	MA.7.AR.2.1
$-144 \leq y$  -144		Recommended Scoring (8 Total Points)	Suggested points are 4 points for solving the inequality, 2 points for the closed circle, and 2 points for the correct ray.
		Additional Notes	Students who reverse the inequality sign on question 4 have a misunderstanding about when to reverse the sign. Students may correctly solve the inequality but draw the ray in the incorrect direction. Questions 1, 2 and 4 will indicated to the teacher if the student has these misconceptions. Students with these misconceptions may benefit from MA.6.AR.2.1 in grade 6 unit 11.
Question	5	Benchmark(s)	MA.7.AR.2.1
5a Inequality $17.50x \leq 112$ 5b $x \leq 6.4$		Recommended Scoring (8 Total Points)	Suggested points is 5 points for correctly writing the inequality and 3 points for correctly solving the inequality that was written. Students should not lose 8 points if they correctly solve an incorrect inequality that they wrote for part A.

Question	6	Benchmark(s)	MA.7.AR.2.2		
6a	<table><tr><td>Equation</td><td>$11(x - 382) = 506$</td></tr></table>	Equation	$11(x - 382) = 506$	Recommended Scoring (8 Total Points)	Suggested points is 5 points for correctly writing the equation and 3 points for correctly solving the equation that was written. Students should not lose 8 points if they correctly solve an incorrect inequality that they wrote for part A.
		Equation	$11(x - 382) = 506$		
Additional Notes	Understanding that a context uses the distributive property is a difficult concept for students. Students who write $11x - 382 = 506$ will get that Esmeralda earns \$80.73 a week. Students who consider their solution in terms of the context will discover their error. This may be helpful for students to be in the habit of doing.				
6b	$x = 428$				
Question	7	Benchmark(s)	MA.7.AR.2.2		
	$w = -34$	Recommended Scoring (8 Total Points)	When determining the distribution of points, teachers should consider the steps that students could use to solve the equation and consider common errors. Students may solve this equation by first dividing both sides by -4 or by distributing the -4 .		

Question	8	Benchmark(s)	MA.7.AR.2.2
	$n = 33.6$	Recommended Scoring (8 Total Points)	When determining the distribution of points, teachers should consider the steps that students could use to solve the equation and consider common errors. Students may solve this equation by first multiplying both sides by -3 or by subtracting 3.8 from both sides.
Question	9	Benchmark(s)	MA.7.AR.2.2
	$4\frac{11}{30} = k$	Recommended Scoring (8 Total Points)	Students could easily lose the coefficient of k , -1 . Consider a deduction of 4 points if a student correctly solves the equation but loses the -1 . Equivalent values of $4\frac{11}{30}$ should be accepted. If students are allowed to use a calculator, then the teacher should set the expectation of rounding.
		Additional Notes	Teachers may want to compare student performance on questions 9 and 10. If students correctly answer 10 but lose -1 on question 9, they may have a misconception on the coefficient being -1 .

Question	10	Benchmark(s)	MA.7.AR.2.2		
$-11.3 = x$		Recommended Scoring (8 Total Points)	When determining the distribution of points, teachers should consider the steps that students could use to solve the equation and consider common errors.		
Question	11	Benchmark(s)	MA.7.AR.2.2		
11a	<table><tr><td>Equation</td><td>$7.80h + 87.40 = 136.15$</td></tr></table>	Equation	$7.80h + 87.40 = 136.15$	Recommended Scoring (8 Total Points)	Suggested points is 4 points for the correct equation and 4 points for the correct solution. Teachers should set the expectation on whether students include the units in their solution.
Equation	$7.80h + 87.40 = 136.15$				
11b	$h = 6.25$ 6.25 hours				
Question	12	Benchmark(s)	MA.7.AR.2.2		
$\boxed{9} (m + \boxed{5.2}) = \boxed{148.5}$		Recommended Scoring (9 Total Points)	Suggested points is 3 points for each correct value that completes the equation.		